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REVIEW

# To draft or not to draft? A systematic review of North American sports' entry draft

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In theory, professional sport "entry drafts" are designed to promote parity by granting poorly performing teams with early selections and winning teams with later selections. While this process has intentions to "level the playing field", mixed findings exist in the literature. The aim of this review is to identify and synthesize the literature examining the efficacy of the draft for professional, North American sport leagues. A systematic review of four databases was performed according to PRISMA (Preferred Reporting Items for Systematic Reviews and Meta-analyses) guidelines. Full-text articles containing relevant data on the draft system for the four major professional North American sports were identified. Further restrictions were made to include articles focusing on a specific outcome regarding future success (i.e., whether the draft related to a measure of future performance). The search returned 10 962 records and after screening, 18 articles were synthesized. Of the articles examined, the measures of future success with relation to draft order were (a) career length and/or number of games played at the majors (n = 8), (b) future performance statistics at the professional level (n = 5), (c) change in winning percentage and/or number of wins produced (n = 3), (d) financial compensation (n = 1), and (e) a combination of measures (a) to (d), (n = 1). Most commonly, the first/early rounds most accurately predicted future measures of success (ie, number of games played, signing bonuses, and playing statistics) across sports. The middle and late rounds were less accurate, with the degree of accuracy increasing slightly in the last rounds. This review highlights several opportunities to better understand the draft process (e.g., potential improvements in middle round picks) and emphasizes the need for more research on analyzing and scrutinizing the draft.

**KEYWORDS**
amateur draft, athlete selection, talent in sport, decision-making, PRISMA, professional sport

## 1 | INTRODUCTION

At many levels of sport performance, making selection decisions (deciding which players will remain in the system and which will leave) is an important part of creating a successful team. In the early stages of competitive sport participation, athletes participate in try-outs, a process of showcasing their skills in hopes of impressing the coach and resulting in their selection to the team. At higher levels of competitive sport, try-outs are conducted on a larger, more robust scale, often comprised of fitness assessments, scouting visits to meet the players, years of performance data, and video footage.[1,2] This information, at least in principle, affords scouts, and other selection personnel the opportunity to make more informed decisions about an athlete's previous performances, his/her current abilities, and future "potential[1]". Not surprisingly, the costs and benefits associated with these decisions can be quite significant. At younger levels of participation, the social and emotional price of telling an athlete that he/she does not have what it takes to be part of the team can be very detrimental.[3-7] At higher levels, professional sports teams accept the risk of making false-positive and false-negative selection errors and wager millions of dollars on the attempt to identify and select the best athletes to their programs in an effort to win the championship title.[2]

It may come as no surprise that access to resources (money, coaches, training, and playing environments) can create financial disparities for teams when trying to select athletes. At the professional level, teams with larger budgets are able to offer larger salaries to more sought-after athletes, often leading to disparity among teams in the league. In an effort to mitigate some of these disparities, the "amateur entry draft" (also known as the "draft") was created. In North America, it was originally introduced in the sport of football in the National Football League (NFL) in 1936 as a way to prevent teams from accumulating "too much talent".[1,8-10] The intentions were, and continue to be, to provide weaker teams a relatively greater opportunity to improve their programs,[11] to deter costly bidding wars for talented young athletes, and help minimize the chance a team would monopolize the best, young players.[12]

Currently, many professional sport leagues (both men's and women's sports) hold entry drafts across many countries. Arguably the most well-known sports in North America with drafts are Major League Baseball (MLB), National Basketball Association (NBA), NFL, and National Hockey League (NHL). These leagues have collective bargaining agreements that govern their actions pertaining to the recruitment and development of athletes.[13] For example, MLB, the NBA, the NFL, and the NHL are unable to recruit players through academies or by purchasing their rights on the transfer market.[14] Rather, athletes are selected only once they have met the minimum league-specific age requirements and are invited to join a team through the entry draft process[2].[14] During the draft, teams take turns selecting (drafting) their future players in a predetermined order which is generally based on the previous year's performance standings (aka the team that comes in last, gets to pick the first athlete, the second last place team gets the second pick, etc.)[3].[12] After the team with the best record selects an athlete, the "first round" of drafting is complete and the processes are repeated until the number of pre-established rounds (based on the league and the year) has been reached.[1] The act of drafting a player means the team earns the rights to sign that athlete to a professional contract. However, drafted athletes typically do not enter to the professional league right away; they are often re-assigned to the teams from which they were drafted, or are asked to join other developmental leagues until the team believes they are ready for the higher level of competitive performance (for more information on the entry draft and its intricacies, please see Boulier et al.,[1] Farah & Baker[14]).

The foundational assumption of the draft is that players selected earlier have more 'potential' than players in the same playing position selected later. Therefore, when executed effectively, there should be a relationship between the order in which players are chosen in the draft and their playing performance in the professional league.[1,15] However, there is both anecdotal support and peer-reviewed evidence questioning the ability of sport executives/team managers to make the most cost-effective selections of athletes to their teams.[1] For example, work by Massey and Thaler[16] suggests the NFL draft choices are "economically inefficient," where managers overvalue early draft choices relative to later draft choices. In a specific case, Tom Brady was chosen in the 6$^{th}$ round, as the 199$^{th}$ pick of the 2000 draft, only to become one of the most renowned NFL quarterbacks in history. Other quarterbacks (such as Warren Moon and David Krieg) had very successful careers in the NFL but were not even selected in the draft.[1] In another example, Staw and Hoang[17] discovered the playing time of NBA players who were drafted early, over-exceeded their playing performance. The authors speculated the managers were unwilling to admit their "errors" and felt obligated to provide playing time to the earlier picks, irrespective of their ability. In contrast to these examples, there are many stories of successful draft decisions. The Pittsburgh Penguins of the NHL, for example, have one of the more notable stories. In 2004, the franchise was reportedly on the verge of folding, but won the draft lottery in 2005 and was awarded the opportunity to select the first athlete in the draft. The team selected

[1]For the sake of the present paper, the term "potential" is used to describe the present demonstration of abilities that are believed to translate into future athletic success.

[2]The MLB is the exception where teams are permitted to sign Latin American players as free agents at 16 years of age. It is also unique in the sense that the MLB has two main drafts and two supplementary drafts each year.

[3]There are exceptions to this rule, as well. To avoid teams purposely performing poorly to gain an advantage in the draft, a lottery system was developed. This lottery system meant that coming in last did not guarantee an earlier pick, but rather increased the odds.

Sidney Crosby, who helped the team appear in 12 playoffs and win the championship in three of those years.[18]

To date, a systematic review using the Preferred Reporting for Information in Systematic Reviews and Meta-Analysis (PRISMA) guidelines[19] has not been conducted exploring the efficacy of the draft for the four major North American professional sports leagues. To help fill this gap in the literature, the purpose of this systematic review is to identify, summarize, and interpret the peer-reviewed literature to date on the efficacy of the draft process in its ability to accurately predict the future performance[4] of draftees. This type of research synthesis is valuable for understanding general trends and conclusions from a research area and intends to fill a gap for a more robust review of the literature on peer-reviewed studies to date according to the criteria outlined below. In addition to academic contributions, this review will have practical implications by outlining which areas of the draft are found to be inefficient, thereby highlighting where more talent identification resources need to be dedicated to improve selection strategies.

## 2 | METHODS

The present systematic review was conducted using the PRISMA guidelines outlined by Moher et al.[19] Using the databases Web Of Science™, SPORTDiscus, EconLit, and JSTOR for Economics and Business, the following search terms were used to examine the abstract and title: "draft" AND "sport," "selection" AND "sport," and "identification" AND "sport." Other databases (e.g., PsychInfo and Scopus) were also searched; however, a small yield of articles combined with a high degree of overlap between articles from the other databases led to the removal of the searches from those databases for the purposes of this review. Additionally, external searches were conducted to yield a greater number of articles outside of these searches (e.g., reading references lists) as per the guidelines from Moher and colleagues.[19]

To determine the specific search terms used in the review, the authors first completed a brief scoping review of the research to identify common language addressing this field of research. The terms "draft," "selection," and "identification" were the most common terms found throughout the scoping review. As the draft is a very specific and unique process, there were very few additional synonyms that existed without the use of the word "draft," which was why only "selection" and "identification" were used as supplementary terms. Once duplicate articles were removed, the refinement process was implemented to eliminate articles with titles and abstracts that did not match the established inclusion (and exclusion) criteria which are detailed below. Articles were kept for further screening if they (a) were in full-text journal article format (ie, conference proceedings, commentaries, or abstracts were not included), (b) published between January 1900 and November 2019, (c) written in English, (d) related to sport, and (e) specifically examining the draft or athlete selection or identification process. Following this initial screening, a secondary screening took place to determine whether the articles met the final inclusion criteria of (i) focused on the four major, North American, men's professional sport leagues (ie, NBA, NFL, NHL, and MLB), and (ii) examining the efficacy of the player draft [5](e.g., how the draft relates to some measure of success in the future). For a figure depicting the PRISMA process and the associated number of articles, please see Figure 1. It is important to mention that under these specific criteria, studies examining undrafted free agents were excluded from our analysis. This is because such players were not selected with a specific pick number; therefore, they do not fit the scope of this study which is focused on *selection number* as a predictor of future performance.

The articles in each league are analyzed by their measure of future success and separated into the following categories (i) length of career/number of games played and likelihood of playing in the majors, (ii) future performance metrics (eg, number of passing yards, number of touch downs, and number of tackles), at the professional level, (iii) financial compensation, (iv) change in teams' winning percentage/number of wins produced, and (v) mixed (some variation of these categories). The following sections will synthesize the articles by each professional sport as well as their respective measure of success along with their context (e.g., draft level, playing position), where applicable (please refer to Table 1 for the overview of articles by sport, measure of success, and findings).

## 3 | RESULTS

Through the data collection phase of the PRISMA, 10 804 articles were extracted from the searching of databases, another 90 were found outside these databases. After screening these articles for duplicates, 8620 articles

[4]Future performance in the context of this paper captures the measurements of an athlete's subsequent playing career in the major leagues following the draft (as opposed to the minor leagues, eg, the AHL), and considers an athlete's playing statistics (eg, number of catches, number of goals), career length, and financial compensation.

[5]This review did not include articles examining re-entry draft or the salary arbitration system.

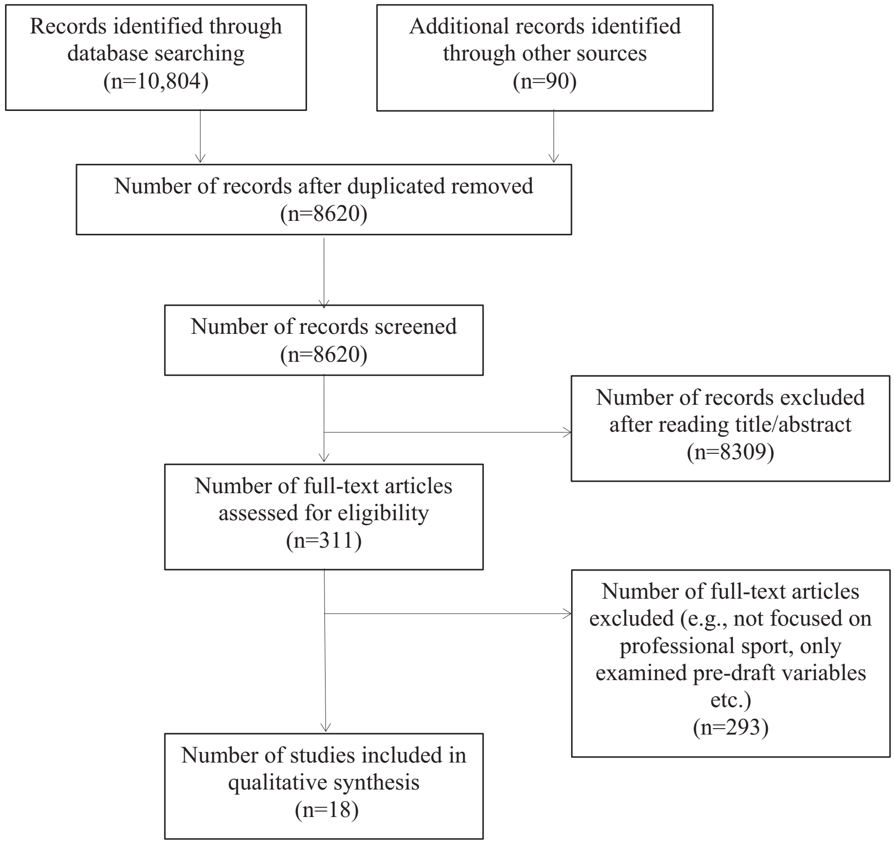


**FIGURE 1** Flow chart of the PRISMA process showing the number of records collected, number of articles screened, and the final number of articles included in the analysis

remained and 8309 of those articles were removed based on their title and abstract. This left a total of 311 full-text articles to be examined in detail, with only 18 meeting the final inclusion criteria as the others either did not focus on professional sport and/or examined only pre-draft variables (e.g., combine scores). Three of the final 18 articles examined the MLB, seven examined the NBA, six examined the NFL, and one examined the NHL. A separate category was made for the sole article that examined all four professional sports.

## 3.1 | MLB

Studies examining the MLB draft were quite sparse (n = 4; 3 with MLB focus and 1 mixed-study), despite the statistically-based movement following the Oakland A's "Moneyball" phenomena.[20] Three articles examined the measure of future success in terms of length of career/number of games played and likelihood playing the majors.[21,23] Two of these found a significant relationship between draft pick and playing time in the majors where the earlier[6] a player was taken in the draft, the increased likelihood of playing in the major leagues[7].[21,23] It is important to note, however, the article by Spurr[23] found a drop-off in the size of the effect after the 34th round out of 40, indicating the draft was more effective in earlier rounds. Conversely, in the solo, mixed-sport study by Koz et al,[22] the authors found no significant differences between the draft rounds for number of games played.

In the remaining study by Burger and Walters,[24] the relationship between draft order and financial compensation was examined. The authors discovered team managers selected lower-quality athletes but did not accurately adjust the signing bonuses to reflect the quality of picks. This pattern was most prevalent in the second and third round. The authors used the term "irrational' to describe team managers" spending approaches.

## 3.2 | NBA

Among the studies of the NBA draft (n = 8; 7 focused on NBA and 1 mixed-sport study), three examined the length of career/number of games played and likelihood of playing in the majors, three examined future performance statistics at the professional level, and one examined a change in winning percentage of teams. Of the

[6]In the articles examining draft order (eg, not articles examining draft round or number of draft picks per team), it is important to note that it is treated as a continuous variable with lower values being more desirable (better) as the player is selected earlier in the draft round.[56]

[7]The study by Caporale and Collier[21], only examined the players in the first round of the draft.

**TABLE 1** Characteristics of the studies included in the review

| Title | Author | Publication year | Sample size | Draft years | Category of study foci | Key findings |
|---|---|---|---|---|---|---|
| *MLB* | | | | | | |
| Scouts versus stats: the impact of Moneyball on the Major League Baseball draft | Caporale and Collier[21] | 2013 | 149 | 1995–1999 | Career length/ games played at the majors | The earlier the player is taken in the draft (in the first round), the greater the likelihood he will make it to the Major Leagues. |
| Uncertain Prospects: Rates of return in the baseball draft | Burger and Walters[24] | 2009 | 2115 | 1990–1997 | Financial compensation | Findings suggest teams are irrational in their spending where they select lower-quality prospects and/or fail to appropriately reduce bonuses in the second and third round. |
| The baseball draft: a study of the ability to find talent | Spurr[23] | 2000 | 2708 | 1966–1968 | Career length/ games played at the majors | The earlier you are taken the draft, the increased likelihood of playing in the majors until later in the draft (round 34) after which draft position has little to no value on the success of a player. |
| *NBA* | | | | | | |
| The dilemma of choosing talent: Michael Jordans are hard to find | Groothuis et al[30] | 2009 | NA | 1987–2003 | Future performance statistics at the professional level | Superstars are more likely to be found early in the draft. However, there are more inaccurate selections than accurate ones in early draft picks |
| The length and success of NBA careers: Does college production predict professional outcomes? | Coates and Oguntimein[25] | 2010 | 290 | 1987–1989 | Career length/ games played at the majors | Early draftees, on average, have longer careers. However, differences in career length are much smaller later in the draft. |
| Predicting success in the National Basketball Association: Stability & potential. | Moxley and Towne[29] | 2015 | 376 | 2001–2006 and 2007–2010 | Future performance statistics at the professional level | Draft order was significantly correlated with future win shares in the first 3 years of players' NBA careers |
| Predictive validity of National Basketball Association draft combine on future performance | Teramoto et al[26] | 2017 | 1092 | 2000–2015 | Career length/ games played at the majors | Draft order was significantly correlated with playing time in the NBA |

(Continues)

**TABLE 1** (Continued)

| Title | Author | Publication year | Sample size | Draft years | Category of study foci | Key findings |
|---|---|---|---|---|---|---|
| Nonlinear judgment analysis: Comparing policy use by those who draft and those who coach | Young[27] | 2008 | 409 | 1989–2004 | Future performance statistics at the professional level | Draft order is a good predictor of future points, assists, rebounds, steals, and turnovers. However, there was no observed effect of draft order on shooting percentage or blocks |
| MoneyRoundball? The drafting of international platers by National Basketball Association teams | Motomura[28] | 2016 | 580 | 1996–2005 | Future performance statistics at the professional level | International players were undervalued in the NBA draft until 2001. However, teams over-adjusted and began to overvalue those players from that point onwards. |
| From College to the pros: predicting the NBA amateur player draft | Berri et al[38] | 2011 | 661 | 1995–2001 | Change in winning percentage/ number of wins produced | Draft order explains very little of the variance in Wins Produced per 48 min (WP48) and total wins produced in the NBA |
| *NFL* | | | | | | |
| Catching a draft: on the process of selecting quarterbacks in the National Football League amateur draft | Berri and Simmons[8] | 2011 | 331 QBs or 1943 observations | 1970–2007 | Future performance statistics at the professional level | Draft pick is not a significant predictor of NFL performance for quarterbacks (based on 19 different performance variables) |
| Deconstructing the Draft: An Evaluation of the NFL Draft as a Predictor of Team Success | Reynolds et al[35] | 2015 | NA | 2000–2010 | Change in winning percentage/ number of wins produced | A moderate, positive correlation was found between draft pick value and change in a team's winning percentage. Additionally, there was a low, positive correlation found between the change in winning percentage and number of first-round draft picks, along with a low, positive correlation between change in winning percentage and total number of picks. |
| Evaluating National Football League draft choices: The passing game | Boulier et al[1] | 2010 | Varies by position | 1974–2005 | Mixed | For both quarterbacks and wide receivers, players taken in earlier rounds have longer careers. Specifically, those taken in the first and second rounds are significantly more successful in their performance (based on multiple performance variables) |

TABLE 1 (Continued)

| Title | Author | Publication year | Sample size | Draft years | Category of study foci | Key findings |
|---|---|---|---|---|---|---|
| The National Football League player draft: does it equalize team strengths? | Lock and Gratz[34] | 1983 | NA | 1970–1981 | Change in winning percentage/ number of wins produced | The correlation between a team's priority position in the draft and its change in performance (based on standings) in future years is low, but statistically significant. |
| Uncertainty, hiring and subsequent performance: the NFL draft | Hendricks et al[32] | 2003 | 5723 | 1979–1992 | Career length/ games played at the majors | Players taken earlier in the draft are more likely to play in the NFL and those with a higher draft position have longer careers. |
| A valuation model for NFL and NHL draft positions | Hurley et al[33] | 2012 | 5611 | 1982–2000 | Career length/ games played at the majors | A significant number of drafted players did not play in the NFL and this fraction tends to grow with draft position. |
| *NHL* | | | | | | |
| The National Hockey League entry draft, 1969–1995: An application of a weighted pool-adjacent-violators algorithm | Dawson and Magee[37] | 2001 | NA | 1969–1995 | Career length/ games played at the majors | The number of NHL games played decreases sharply as the draft pick number increases, but flattens out in the later picks. |
| *Mixed Sport* | | | | | | |
| Accuracy of professional sports drafts in predicting career potential | Koz et al[22] | 2012 | 1028 | 1980–1989 | Career length/ games played at the majors | NHL career length significantly differs across draft rounds in a descending fashion, but stabilizes in the later rounds of the draft. |
| | | | 407 | 1980–1999 | Career length/ games played at the majors | In the MLB, there are no significant differences between the draft rounds for games played |
| | | | 2380 | 1980–1999 | Career length/ games played at the majors | In the NFL, Players drafted earlier played more games than those drafted later. |
| | | | 407 | 1980–1999 | Career length/ games played at the majors | In the NBA, there were no significant differences between the draft rounds for games played |

Abbreviations: MLB, Major League Baseball; NBA, National Basketball Association; NFL, National Football League; NHL, National Hockey League.

three articles studying length of career/number of games played and likelihood of playing in the majors, Coates and Oguntimein[25] found early draftees, on average, had longer careers—however, differences were much smaller later in the draft, suggesting the teams' abilities to identify future performance is more effective earlier in the selection process. Koz et al[22] found no significant differences between draft rounds for games played, whereas Teramoto et al[26] found draft order was significantly correlated with playing time in the first 3 years of a player's NBA career.

Four studies investigated future performance statistics at the professional level. Young[27] found the earlier draftees were selected, the more points, assists, rebounds and blocks they scored, and the fewer steals and turnovers they had. However, there were no observed effects of draft order on shooting percentage or blocks. Motomura[28] examined the specific relationship of NBA efficiency (NBAEFF; the net accumulation of positive and negative recorded basketball "acts" (points rebounds, assists, and steals), and WP (wins produced), which together were considered a measure of efficiency. With respect to draft position, when comparing international players to players who are from the USA, Motomura[28] found international players drafted through 2001 tended to outperform their expectations (adjusted for draft positions), which led to teams subsequently drafting more international players. However, after 2001, first-round picks tended to underperform, implying that teams overcompensated. This further suggests teams have not reached optimal evaluation of international players relative to USA-trained players. In the study by Moxley and Towne,[29] the authors found that draft order was significantly correlated with NBA win shares during the first 3 years of players' careers. This win share metric uses a mixture of efficiency measures (eg, how often players are involved in plays) and while not a perfect measure of success, it is believed to be a reliable metric of "all round" performance. In the final study in this category, Groothius et al[30] examined the efficiency of draft picks based on an efficiency formula (combining points, rebounds, assists, steals blocks, field goal attempts, field goals made, free throws attempted, free throws made, and turnovers). They found more false positives existed than correct decisions with draft picks taken higher in the draft. That said, if "superstar players" were found, they were usually identified early. The authors noted "the dilemma of choosing talent is not so much a winner's curse but more like a purchase of a lottery ticket. Most times you lose, but, if you are going to win, you must buy a ticket" (pp. 3198).

Finally, Berri et al[31] examined "wins produced" finding no convincing evidence that draft order predicts future NBA wins. More specifically, they found draft order only explained <5% of wins produced per 48 min of play, and 7% of total wins produced.

## 3.3 | NFL

In the peer-reviewed literature, the NFL tied the NBA for the greatest number of studies (n = 7; 6 focused on NHL and 1 mixed-sport study). Within this sub-set of articles, three examined length of careers and or/number of games played at the professional level,[22,32,33] two studies examined the change in future winning percentages,[34,35] one examined players' expected salaries,[36] one study examined future performance statistics,[8] and one study examined multiple indicators and were therefore considered "mixed".

Among the studies examining length of careers and or/number of games played, Hendricks et al[32] and Koz et al[22] both found significant relationships between draft order and games played. Koz and colleagues[22] revealed earlier rounds were more significantly related to career length than later rounds, once again reinforcing the notion that the draft is more effective in the earlier rounds. Conversely, Hurley et al[33] found a significant number of the players drafted did not play in the NFL and this fraction tended to grow with draft position.

For studies of winning percentages with relation to draft order, Lock and Gratz[34] found a low, but statistically significant relationship between teams' positioning in the draft (i.e., how early they select) and their change in winning percentage. The authors suggested "the benefits derived from the current draft in terms of the equalization of team strengths are substantial enough to justify these heavy costs" (pp. 26). Similarly, Reynolds et al[35] found a statistically significant, moderate-sized effect between draft pick value and a team's winning percentage, and a low, but statistically significant relationship between the total number of first-round picks and winning percentage. The authors suggested their results indicated the draft was effective in its ability to contribute to "competitive balance".

In their study, Berri and Simmons[8] found that quarterbacks selected early in the draft have better per-game metrics. However, when per-play metrics were analyzed, they found that quarterbacks selected early perform worse on a per-play basis than ones selected later. What this suggests is that early draftees simply play more, but do not perform better than their counterparts selected after them. This conclusion was based on 19 variables (e.g., passing yards per pass attempts, touchdowns per pass attempt, and touchdowns per pass attempt) across all pick positions.

Last, in the mixed-focus study by Boulier et al,[1] the authors found both quarterbacks and wide receivers taken in earlier rounds had longer careers—with those taken in the first and second rounds being significantly more successful than quarterbacks and wide receivers selected after the second round. For quarterbacks, the number of passes thrown declined significantly from the top of the second round to the top of the third round. Additionally, the earlier a quarterback was drafted, the better his quarterback rating (indicator of performance), and for this measure, there was a sharp drop from the first to the second round. Similarly, for wide receivers, an earlier draft pick was related to more passing yard receptions.

### 3.4 | NHL

As noted, there was only one peer-reviewed study dedicated solely to examining the NHL draft,[37] as well as one that included the NHL in their mixed-sport approach.[22] Both of these articles examined length of career/number of games played and likelihood of playing in the NHL as the measure of future success. Dawson and Magee[37] found the number of NHL games played decreased sharply as the draft pick number increased, but the relationship flattened out in later picks. Similarly, Koz and colleagues[22] found NHL career length significantly differed across draft rounds in a descending fashion, but stabilized in the later rounds of the draft. These findings speak, once again, to the earlier rounds of the draft showing higher accuracy than the later rounds.

## 4 | DISCUSSION

Of the 18 studies (from an initial search result of 10 804 records) examining the efficacy of the draft in predicting a future measure of success for the four, major North American professional leagues, most (n = 8) considered length of career/number of games played and likelihood of playing in the majors as their key indicator of "success". Further, five others considered future performance statistics at the professional level, three examined a change in winning percentage/number of wins produced, while the remaining studies considered indicators of financial compensation (n = 2) or a mixture of outcomes (n = 1). Overall, the studies in this systematic review indicate the draft across all four leagues has inefficiencies. While some demonstrated the strength of the draft process,[34,35] many authors questioned the ability of the draft to accurately identify "talent" and for those in decision-making positions to act in a rational and logical manner.[8,22,24,30,38] Particularly, the general findings indicate that first-round draftees do, in fact, go on to outperform their peers in the future. However, the large discrepancy in future performance in subsequent rounds, which form the majority of the draft, suggests decision makers' abilities to accurately find talent past the initial rounds is suboptimal. That said, the diversity in the study foci combined with the overall sparse number of articles make it difficult to determine clear conclusions about the efficacy of the player draft in relation to measures of success. However, in the following discussion, we will attempt to draw some inferences from the research summarized.

### 4.1 | Imperfect data for forecasting

Predicting the future, in most domains, is a highly challenging task. Even in a data-rich environment like the North American professional sport leagues, forecasting an athlete's future performance remains a largely inaccurate practice. One plausible explanation for this inaccuracy is there is no single formula for being "talented". While there are certain sports that have more uniform physical profiles for expert performers (e.g., rowing and swimming), many sports, including the professional sports in this review, have athletes with a high degree of diversity in the genetic predispositions and developmental experiences, as well as the physiological and psychological makeups of its performers. An added challenge occurs when talent scouts/coaches are required to make judgements about an athlete's potential (who to select and who to de-select) during all stages of the elite sport pathway—many of which are during stages of rapid change (eg, maturation).[39]

At earlier stages of an athlete's life, there are many moving parts including physical, social, and psychological systems developing at different rates. Even at the professional level of sport, when physical and physiological changes are more stable, the ability to accurately capture an athlete's abilities and skills using standardized tests is questionable. Multiple studies have examined the validity of the college/university performance tests and have even suggested the money allocated for formal assessments (often in the structure of a "combine") is not wisely spent.[2,9] The combine is often used prior to the draft and is usually a multi-day series of tests designed to assess an athlete's physicality (and in some leagues, psychology). While these events started as data collection opportunities, they have evolved into major media events, and coverage brings a significant amount of publicity to the leagues along with ticket sales and marketing revenues.[34] Despite their popularity, however, multiple studies show combine-type events lack a meaningful degree of predictive validity,[2,40] despite being a relatively important pre-cursor for draft position.[8]

It should be noted scouts develop extensive player profiles that capture information extending beyond what is extracted from the combine data.[2] For example, scouts and other staff perform multiple player interviews to better understand the athlete on a more personal level and to gain insight into his "soft" skills.[1] These questions may be related to the athlete's upbringing, family dynamics, sport background, leadership opportunities, times of resiliency/adversity among other factors, and may provide the team with a better picture of whether or not the player is a good "fit" to the team. While the interview process has been questioned regarding its validity and reliability (e.g., it can be difficult to measure less-tangible skills), the role of the scout is often celebrated for its ability to glean information about an athlete.[41] How this information is used in the final decision-making process and how it is translated to the other departments (i.e., analytics) is not very well understood and is an important area for further exploration.

### 4.2 | Cognitive errors and biases in decision-making

There is considerable economic risk with the drafting process and starting salaries vary dramatically between players (e.g., the first draft pick's contract could be up to four times the amount of the last pick in the first round).[16] Previous research has shown; however, that managers of professional teams significantly have over-valued early draft choices over late draft choices without solid economic rationale.[16,22] Tied closely with imperfect data for decision-making is the fallibility of the judge/scout when using the information they acquire. There are many steps in the decision-making process (eg, watch games, administer physical and mental tests, and conduct personal interviews) before a draft decision is made, leaving room for error in analysis and judgment. As noted in other domains (eg, medicine, economics, and behavioral psychology), humans are prone to many cognitive biases and errors when it comes to making decisions under uncertainty.[42,43] One of the more well-researched biases in the sport domain is relative age bias, where coaches and selectors have a tendency to select relatively older athletes because of their likelihood to appear superior compared to their relatively younger counterparts.[44] In another example, people irrationally overvalue something they own (object, asset, etc.) beyond its true value, making them more likely to retain it (called the endowment effect). In the context of talent identification, the idea that a scout took the time to get to know a player on a more personal level, track his performance over time, meet his parents and have developed a relationship with him and his family, may lead the scout to overvalue the player and blind him from seeing his faults. A related issue is the "halo effect", which reflects the potential for a few positive qualities/characteristics to overshadow a player's negative ones, leading to a more positive view of the athlete (or vice versa). As these effects imply, there can be multiple factors at play (either consciously or sub-consciously) that influence evaluator's decision-making process. While some of these cognitive shortcuts can be time and energy saving and lead to correct outcomes, many can lead to predictable and systematic errors (for more information on the cognitive errors in decision-making, see Hastie & Dawes[42]; Tversky & Kahneman[45]). These errors are further complicated by the decision-making conditions where unclear motives, political agendas, and different viewpoints add to the complexity and accuracy of the decision-making process.

### 4.3 | Social influences

In addition to selecting a player based on his potential to improve the team, selection decisions can also be swayed by politics and other motives. This includes decisions around which athlete has a lower "risk" in terms of liability in an effort to save public image, or which athlete is the most marketable to fans to increase profit from ticket and merchandise sales. As viewership and other fan-related concerns are important considerations for a team, it is plausible some draft decisions could be influenced by the intention to "please the crowd". For instance, Treme and Allen[15] tested whether talent and popularity explained the emergence of rookie wide receivers in the NFL in the 2001–2006 seasons. The authors found evidence that a player's media exposure prior to the draft helped explain the variation in both draft rank and first-year salary.

Similarly, it is plausible teams will negate their statistical models to select an athlete that has been touted by the media and other pundits. Take, for example, a team trying to make a selection decision for the draft that sees their own models rank a certain player higher than sports pundits and media. It would take courage for a team to knowingly go against the media and potentially expose themselves to backlash. It is, therefore, likely harder for a team to make a decision that aligns with the media and for it to fall short, but when a team makes decisions that goes against the media and has a failing outcome, it becomes "obvious" where the errors

[8]For a counterargument, see Teramoto et al[26] on the predictive validity of national basketball association draft combine on future performance and Tarter et al[55] on the use of aggregate fitness indicators to predict transition into the national hockey league

are, even though they may be unrelated. It is not out of the question, then, for a team to make a decision that takes the "path of least resistance" in these situations. Whether this is conscious or sub-conscious, taking short cuts in the decision-making process is a common and even probable occurrence. With the exception of Treme and Allen,[15] however, evidence (beyond anecdotal support) is quite sparse. This speaks to the need for research investigating such influences in the context of athlete selection.

### 4.4 | Influences related to career and development trajectories

In addition to the abovementioned factors affecting accurate draft selections, there is a host of other reasons why a player later emerges as a "non-successful pick". One of those reasons is an early career injury that halts the athlete's development. While the field of sports medicine has uncovered a great deal about injury types that affect career trajectories,[46-49] there remains a scarcity of studies that consider injury as a confounding variable when assessing draft accuracy. Conducting more of such research, whether in the peer-review system or internally by teams, may lead to a better understanding of the factors underlying the success of draft selections.

Another factor to consider is the relative strengths of teams to which athletes are selected. Given that the draft is conducted in reverse order of the previous year's standings, early draftees are typically selected into poorly performing teams that may not have a strong supporting cast for the draftee to perform well. Not to mention that teams in certain leagues—such as MLB—use a player rotation system wherein a draftee may not receive as many playing opportunities as his peers, leading to decreased performance over time.[22]

### 4.5 | Inherent challenges of chaotic systems

Perhaps the most glaring challenge to making accurate selection predictions is the dynamical and complex nature of human performance. It may come as no surprise that elite sport performance is the product of the complex interconnectedness of the athlete's genetics, his/her upbringing, along with the environment and systems in which he/she operates within. This, in itself, makes sport (at least in part) inherently chaotic and random. Despite best scientific efforts and technological advancements, sport performance may never truly be calculable and/or predictable and it will likely mean making perfect selections will never be feasible. That being said, the emphasis should not be on making perfect selections, but rather shifting the focus on finding ways to be less incorrect. For this reason, it is important to consider what is deemed "acceptable" when it comes to selection accuracy at various steps along the sport pathway.

## 5 | LIMITATIONS AND FUTURE DIRECTIONS

Despite the potential implications of this systematic review, there are limitations to consider. One of the most notable is this review only reflects information published through the peer-reviewed system, which is not necessarily indicative of the current climate of knowledge that exists within the field. Much of the accessible information is shared via journalism, online blogs, podcasts, and other media platforms. Additionally, it is well understood there is a lack of "sharable knowledge" that exists in professional sport. Often, findings are not made public through peer-reviewed literature in order to keep knowledge "in-house" for a competitive advantage. Another possible reason for the lack of peer-reviewed work in professional sport may be related to the time needed for the peer-review process to occur. For example, longitudinal and even cross-sectional research can take months or years to publish, and in an industry that moves rapidly, many organizations cannot wait for the peer-review process.

It is noteworthy that most papers in this systematic review relied on data from 1970 to 1999 or early 2000s, which may not capture the contemporary strategies teams currently have in place when drafting. For instance, the recent use of artificial intelligence and automation has resulted in more advanced capturing of players' pre-draft performance and analytics,[50-53] meaning teams are now able to utilize data of higher quality when making draft decisions. Similarly, advancements in the fields of physiology and sport psychology may also have aided in making better talent evaluations at the draft.[54,55]

Additionally, the fact that very few studies within the same sport examined the same measures of success, and the added complexity of comparing different types of statistical analyses and reporting practices, created a challenge to presenting quantitative findings in a logical and comprehensible way. Ideally, the present study would have reported the statistical relationships between the variables examined corresponding each playing position, but due to inconsistencies and lack of standardization across articles, combined with the sheer number of variables examined in some articles, the authors were unable to do so.

In spite of these limitations, this systematic review highlights the clear need for research on the efficacy of the draft process across the four major leagues. This is especially true for the MLB and the NHL which have very few articles relating to the efficacy of the draft in predicting future measures of success. Even using the articles examined, there are no definitive results that show the draft is effective in its current structure; therefore, more research is needed to determine the inefficiencies and, subsequently, approaches for improvement.

Additionally, future directions for research include examining the differences between international draft programs in terms of their processes and accuracy rates. This could perhaps draw attention to the unique ways "talent" is identified, selected for, and developed. Highlighting these differences (and perhaps similarities) may, in turn, help to create opportunities for knowledge sharing and provide support for policy and procedure (re)considerations. Moreover, it would be beneficial to examine the athlete selection process from an even broader perspective by assessing free agencies to better understand the (in)efficiencies of the draft system at large.

## 6 | PERSPECTIVE

In all four sports, from the studies comparing draft order by round, it was most commonly found that the first round most accurately predicted a measure of future success. As the rounds progressed, however, the efficacy of that variable to predict future success decreased.[22,32] In short, teams and their managers/executives have success in the early and late rounds of the draft, but the middle rounds are less accurate. These findings have implications for those involved in the drafting process, suggesting there is insufficient evidence to show the draft is a truly effective method for predicting certain measures of success.

In addition to the questionable effectiveness of predicting future success, the present systematic review highlights the limited existing evidence to support the notion that the reverse order of the draft promotes equality, which is a fundamental principle of the draft system.[34] Based on the present findings (and with so few articles in each sport category), the authors cannot draw any conclusion with confidence that athletes selected earlier in the draft (from weaker teams) are significantly better than those chosen later in the draft (by stronger teams).

In some ways, it is surprising that the efficacy of the draft has not received more scientific scrutiny. In other ways, it is understandable based on the proprietary nature of the information and how it can act as a competitive advantage for teams. Overall, the firmest conclusion we can make is that more research is needed to be able to truly test the efficacy of the draft to better understand its strengths and weaknesses.

## CONFLICTS OF INTEREST

The authors do not have any conflicts of interest to report.

## AUTHOR CONTRIBUTIONS

Kathryn Johnston, Lou Farah, and Joseph Baker contributed equally to the conception and design. Kathryn Johnston and Lou Farah performed material preparation, data collection, and analysis. Kathryn Johnston, Lou Farah, and Harleen Ghuman performed coding and coding reliability. Kathryn Johnston wrote the first draft of the manuscript. Lou Farah and Joseph Baker provided feedback. All authors read and approved the final manuscript.

## DATA AVAILABILITY STATEMENT

The data that support the findings of this study are available from the corresponding author upon reasonable request.

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